MIL-S-81775B(AS)
6 December 1974
Superseding
MIL-S-81775A(AS)
30 November 1973

MILITARY SPECIFICATION
SERVICE SET, HYDRAULIC SYSTEM
MX-9043( )/AWM-23 (V)

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the performance, design, and test requirements for the Hydraulic System Service Set (HSSS) MX-9043( )/AWM-23 (V).

2. APPLICABLE DOCUMENTS

2.1 General. The following documents, of the issue specified in AS-4053 Applicable document List for AWCS Test Set AN/AWM-23(V), form a part of this specification to the extent specified herein, MIL-T-81734(AS) is the applicable general specification for the HSSS described herein (see 6.6).

SPECIFICATION

Military

MIL-M-7793	Meter, Time Totalizing
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for
MIL-T-81734(AS)	Test Set, Airborne Weapon Control System AR/AWM-23 (V)

STANDARDS

MIL-STD-761	Electrical Power, Alternating Current for Shipboard Use, Characteristics and Utilization of

PUBLICATIONS

Hughes Aircraft Company

3282212	Acceptance Test Procedure Hydraulic System Service Set

2.2 Availability of documents. -Applicable (see 6.6).

2.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary documents are in confllict, the following

FSC 4935

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precedence shall apply:

a, Contract. The contract shall have precedence over any other document.

b. This specification. This specification shall have precedence over all applicable subsidiary documents (see 6.6).

c. Referenced documents. Any document referenced in this specification shall have precedence over all applicable documents referenced therein.

3. REQUIREMENTS

3.1 Materials, parts, and processes. Applicable (see 6. 6).

3.2 Design and construction. Applicable except as follows

3.2.1 Design features. Applicable except as follows.

3.2.1.1 Method of operation. Applicable.

3.2.1.2 Commercial test equipment. Not applicable.

3.2.1.2.1 Other commercial components. Applicable.

3.2.1.3 Circuit grounds. Applicable.

3.2.1.4 Interchangeability. Applicable.

3.2.1.5 Identification of product. Applicable.

3.2.1.6 Human factors. Applicable.

3.2.1.7 Test set generated acoustic noise. Applicable.

3.2.1.8 Forced-liquid cooling. Not applicable.

3.2.1.9 Space and weight limitations. The maximum dimensions of the HSSS, including the cover, shall be 22-1/2 inches by 16-1/2 inches by 11-5/8 inches. The maximum weight of the HSSS shall be 55 pounds.

3.2.1.10 Connectors. Applicable.

3.2.1.11 Secondary electrical power. Not applicable.

3.2.1.12 Interface. The HSSS shall provide the following interfaces.

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3.2.1.12.1 WRA. The HSSS shall interface with Antenna AS-2764( )/ AWG-9 through two hydraulic connections.

3.2.1.12.2 Electrical power. The HSSS shall interface with the electrical power of 3.2.2.1 through the cable of 3.6.14.

3.2.2 Utilities. Applicable.

3.2.2.1 Electrical power. Electrical power required for operation of the HSSS shall be not more than 300-volt-amperes of 115-volt, 400 Hertz (Hz), single phase power. This alternating current (ac) power form shall conform to MIL-STD-761, Type I.

3.2.2.2 Clean air. Not applicable.

3.2.2.3 Potable water. Not applicable.

3.2.2.4 Liquid coolant. Not applicable.

3.2.2.5 Helium. Not applicable.

3.2.2.6 Nitrogen. Not applicable.

3.2.2.7 Hydraulic fluid. Applicable.

3.2.3. Service conditions. Applicable.

3.2.4 Maintainability. Applicable.

3.2.5 Reliability. Applicable. The mean time between failures (MTBF) shall be 500 hours minimum.

3.2.6 Electromagnetic interference (EMI). Not applicable.

3.2.7 Utilities protection. Applicable.

3.2.7.1 Liquid cooling protection. Not applicable.

3.2.7.2 Clean air protection. Not applicable.

3.2.7.3 Electrical power protection. Applicable.

3.2.8 Calibration cycle. Applicable. Calibration requirements and requirements for support equipment to maintain the HSSS will be stated in the Test Set Maintenance Engineering Analysis (MEA). Procedures will be stated in the Technical Manuals.

3.2.9 Fluid capacity and filtration. The HSSS shall provide storage for one gallon of hydraulic fluid and shall provide fluid filtration of 3 microns absolute.

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3.3.1 General. The HSSS shall provide for hydraulic fluid refilling, purging, bleeding and filtering of Antenna AS-2764( )/AWG-9. The HSSS shall maintain an operating pressure of 35 ± 5 pounds per square inch gage (psig) with a minimum rate of 0.5 gallon per minute (gpm).

capacity of at least one gallon of hydraulic fluid.

3.6.2 Hydraulic fluid level gage. The hydraulic fluid level gage shall be a float gage capable of indicating the relative fullness of the hydraulic reservoir.

3.6.3 Hydraulic pressure gage. The hydraulic pressure gage shall have a range of zero to 200 psig pressure at the output of the filter assembly.

3.6.4 Hydraulic pressure regulator. The hydraulic pressure regulator shall be manually adjustable to control the output pressure of the HSSS to the pressure specified in 3.3.1.

3. 6.5 Hydraulic supply shut-off valve. The hydraulic supply shut-off valve shall be a manually operated panel mounted valve to provide control of the hydraulic fluid flow rate to the radar antenna hydraulic system.

3.6.6 Hydraulic return shut-off valve. The hydraulic return shut-off valve shall be a manually operated panel mounted valve to central the hydraulic fluid returned from the radar antenna hydraulic system.

3.6.7 Hydraulic check valve. The hydraulic check valve in the supply line shall prevent the back flow of hydraulic fluid into the HSSS in the event of power failure in the HSSS.

3.6.8 Hydraulic return line sight glass. A sight glass shall be provided in the hydraulic return line to permit visual inspection for bubbles in the hydraulic fluid.

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3.6.9 Three micron absolute filter. A three micron absolute full flow filter shall provide filtration of the hydraulic fluid.

3.6.10 Differential pressure indicator. The differential pressure indicator shall indicate, by a visual pop-up indication, when pressure drop due to contamination of the filter of 3.6.9 exceeds 80 ± 12 psi.

3.6.11 Flexible hoses. Two flexible hoses shrill be provided, each 10 ± 1 feet in length. The hoses shall have quick-disconnect fittings to mate with the radar antenna hydraulic system.

3.6.12 Hydraulic pump switch. The hydraulic pump switch shall be a panel mounted toggle switch and shall control electrical power to the HSSS.

3.6.13 Elapsed time indicator. An elapsed time meter conforming to MIL-M-7793 shall be provided. The meter shall be readily visible during operation of the HSSS and shall be capable of registering at least 9,999 hours.

3.6.14 Power cable. A power cable shrill be provided in accordance with MILT-21200. The connector shall be MS3107A-16S-8P with MS3057-8A cable clamp. The connector shall have 115 volts, 400 Hz single phase power on Pin A with the return on Pin D. The length of the cable shall be 25 ± 0.5 feet.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Applicable.

4.2 General. Applicable.

4.2.1 Classification. Applicable.

4.2.2 Individual items. Applicable.

4.2.3 In-process inspection. Applicable.

4.2.4 In-process data. Applicable.

4.2. 5 Acceptance test plan. Applicable. Table I shall apply.

4.2.6 Test results. Applicable.

4.2.7 Acceptance. Applicable.

4.2.8 Utilities. Applicable.

4.2.9 Configuration. Applicable.

4.3 Tests

4.3.1 Electrical and electronic items - operational run-in test. Applicable.

4.3.2 Mechanical items. Applicable.

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TABLE I

ACCEPTANCE TEST PLAN FOR HYDRAULIC SYSTEM SERVICE SET

MX-9043( )/AWM-23(V)

1.0 TEST OBJECTIVE. To demonstrate functional operation of the Hydraulic System Service Set.

2.0 TEST CONDITIONS

a. Test Equipment

(1) HSSS Console Test Position (HAC Part No. T493703-61-3)

b. Utilities Required (information only, ref. 4.2.8)

(1) Power as specified by 3. 2.2.1.

3.0 TESTS (In accordance with Acceptance Test Procedure Document, 3282212)

a. Operational run-in tests arc not required.

b. Perform flow and pressure tests in accordance with the acceptance test procedures.

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4.4 Environmental test conditions. Applicable.

4.5 Test location. Applicable.

4.6 Acceptance schedule. Applicable.

4.7 Test procedures. Applicable.

4.8 Rejection and retest. Applicable.

5. PREPARATION FOR DELIVERY

5.1 General. Applicable.

6. NOTES

6.1 Intended use. The HSSS is intended for use as part of the AN/AWM-23 Test Set to fill and bleed the AN/AWG-9 radar antenna hydraulic system at the organizational and intermediate levels of maintenance.

6.2 Ordering data. Applicable.

6.3 Definitions. Applicable.

6.4 Nomenclature cross reference. The contractor part number for Service Set, Hydraulic System MX-9043( )/AWM-23(V) is 493703-1XX. The constructor part number suffix is controlled by the configuration list in the contract.

6.5 Liquid cooler. Not applicable.

6. 6 Use of this specification. This specification is a detail specification which uses MIL-T-81734(AS) as a general specification. Where the corresponding paragraphs of MIL-S-81775(AS) are the same as MIL-T-81734(AS) the word "applicable" is used in this specification. When the term "not applicable" is used in this specification. it means that the corresponding paragraph requirements of MIL-T-8l734(AS) are not required to be met (see 6.6 of MIL-T-81734(AS)).

6.7 Abbreviations and acronyms. Applicable. In addition the following shall apply.

ac	alternating current
HSSS	Hydraulic System Semite Set
MEA	Maintenance Engineering Analysis

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-81775B(AS) SERVICE SET, HYDRAULIC SYSTEM MX-9043( )/AWM-23(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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